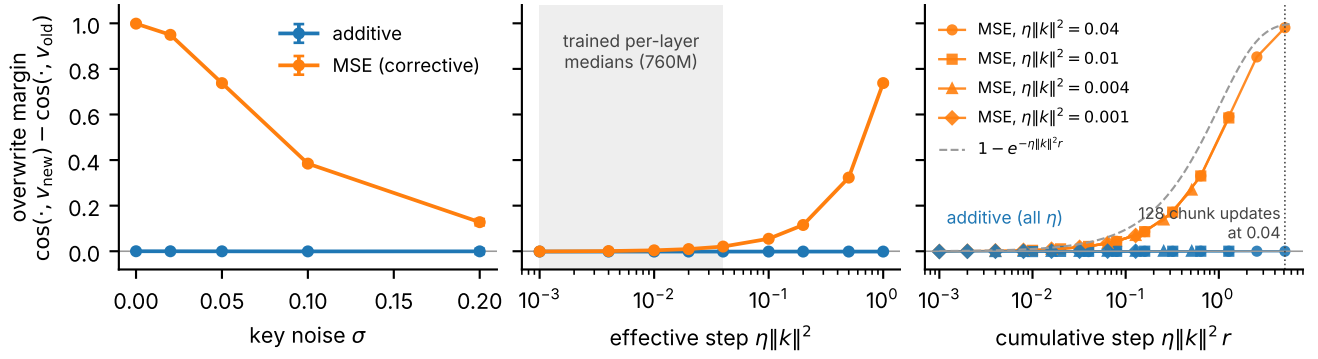
(a) one write each, $\eta\|k\|^2 = 1$
overwrite margin
$\cos(\cdot, v_{new}) - \cos(\cdot, v_{old})$
1.0
0.8
0.6
0.4
0.2
0.0
additive
MSE (corrective)
0.00
0.05
0.10
0.15
0.20
key noise σ
(b) one write each, noise 0.05
trained per-layer
medians (760M)
10^{-3}
10^{-2}
10^{-1}
10^{0}
effective step $\eta\|k\|^2$
(c) r writes each, noise 0.05
MSE, $\eta\|k\|^2 = 0.04$
MSE, $\eta\|k\|^2 = 0.01$
MSE, $\eta\|k\|^2 = 0.004$
MSE, $\eta\|k\|^2 = 0.001$
$1 - e^{-\eta\|k\|^2 r}$
additive (all η)
128 chunk updates
at 0.04
10^{-3}
10^{-2}
10^{-1}
10^{0}
cumulative step $\eta\|k\|^2 r$